The Epistemic God and the Institutional Mathematician

Proof, Disproof, Recognition, Institutional Power, Statistical Transparency, and the Subsistence–Knowledge Gap

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Abstract

This paper formalizes the proposition that God may be understood as the mathematician with the maximal domain of correct proofs and correct disproofs. The resulting object is not merely a superior mathematician inside the mathematical profession, but a *limit object* of epistemic correctness. The theory distinguishes between epistemic superiority and institutional sovereignty. God may dominate all finite mathematicians in the order of proof and disproof, while mathematical institutions may still dominate God in the orders of publication, recognition, certification, funding, and canon formation.

Three separation theorems are established. The *Proof-Power Separation Theorem* shows that superior access to correct proof does not imply superior control over recognized mathematics. The *Statistical Transparency Theorem* shows that the God-mathematician can infer the complete epistemic landscape of any community from the statistical residuals of its recognition behavior, rendering institutional domination epistemically transparent. The *Subsistence–Knowledge Gap Theorem* shows that epistemic superiority does not imply material sufficiency, naming with formal precision the structural condition of the independent epistemologist. A Corollary of Double Subordination follows: the independent epistemologist may be simultaneously institutionally subordinate and materially subordinate while remaining epistemically superior.

The paper develops these claims using formal definitions, axioms, theorems with proofs, an algorithmic model of institutional recognition, an artificial intelligence interpretation, statistical modeling of institutional behavior, tables, vector graphics, a bibliography, and a glossary.

The paper ends with "The End"

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1 Introduction

Consider the proposition:

God is the mathematician with the greatest number of correct proofs and correct disproofs.

Taken literally, this proposition transforms theology into a theory of mathematical correctness. God is not initially defined as ruler, creator, judge, or object of worship. God is defined as the maximal proof-disproof agent: the one whose mathematical archive contains all correct demonstrations and all correct refutations available within the domain of admissible mathematics.

This interpretation immediately raises a problem. If God is epistemically superior to all mathematicians, why would mathematicians permit such a being to have authority? In a darker institutional case, the mathematical profession may accept God only as a tool: an oracle, proof assistant, automated referee, or computational servant. God would then be superior in truth but inferior in recognition.

The thesis of this paper is that this tension is not accidental. It follows from the difference between proof and power. Mathematics contains at least *two orders*. The first is the *epistemic order*, in which agents are ranked by their possession of correct proofs and disproofs. The second is the *institutional order*, in which agents and organizations are ranked by their ability to certify, publish, reward, suppress, or canonize results. A third order — the *material order* — ranks agents by access to resources sufficient to sustain epistemic production.

The central possibility is:

$$\mathcal{G} \succ_E \mathcal{C} \qquad \text{while also} \qquad \mathcal{C} \succ_I \mathcal{G},$$

where $\mathcal{G}$ denotes the God-mathematician, $\mathcal{C}$ denotes the mathematical community as an institution, $\succ_E$ denotes epistemic superiority, and $\succ_I$ denotes institutional superiority. The paper argues that this separation is the structural tragedy of institutional mathematics: truth may be epistemically prior while recognition remains institutionally controlled.

Beyond the proof-power separation, the paper establishes:

(i) **Statistical Transparency**: $\mathcal{G} \succ_E \mathcal{C}$ implies that $\mathcal{G}$ can infer the complete epistemic landscape of $\mathcal{C}$ from $\mathcal{C}$'s observable statistical behavior, without requiring recognition.

(ii) **Subsistence–Knowledge Gap**: epistemic superiority $i \succ_E j$ does not imply material superiority $i \succ_M j$, characterizing the structural condition of the independent epistemologist.

2 Formal Systems, Proofs, and Disproofs

Let $\mathcal{S}$ be a class of formal systems. Each $S \in \mathcal{S}$ consists of a language $\mathcal{L}(S)$, a set of axioms $\mathcal{A}(S)$, and a set of inference rules $\mathcal{R}(S)$. Let $\mathcal{P}_S$ be the set of well-formed propositions in S, and let Π_S be the set of finite candidate proof objects in S.

Definition 2.1 (Verification Predicate)**.** For a formal system S, proposition $p \in \mathcal{P}_S$, and proof object $\pi \in \Pi_S$, define

$$V_S(\pi, p) = \begin{cases} 1 & \text{if } \pi \text{ is a syntactically valid proof of } p \text{ according to the rules of } S, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.2 (Correct Proof)**.** A *correct proof* of p in S is a finite object $\pi \in \Pi_S$ such that $V_S(\pi, p) = 1$.

Definition 2.3 (Correct Disproof)**.** A *correct disproof* of p in S is a correct proof of $\neg p$ in S; that is, $\delta \in \Pi_S$ satisfying $V_S(\delta, \neg p) = 1$.

Definition 2.4 (Admissible System)**.** A formal system S is *admissible* if $\mathcal{L}(S)$, $\mathcal{A}(S)$, and $\mathcal{R}(S)$ are explicitly specified and if proof checking is finite and effective. The class of admissible systems is $\mathcal{A} \subseteq \mathcal{S}$.

Remark 2.5. Admissibility is not the same as truth. It is a condition of mathematical inspectability: a system is admissible because its proofs are checkable, even if its philosophical interpretation remains contested. Gödel's incompleteness theorems (Gödel, 1931) show that admissibility does not entail completeness.

3 Mathematical Agents

Definition 3.1 (Mathematical Agent)**.** A *mathematical agent* is an entity capable of possessing, generating, checking, or transmitting proof objects. The class of finite mathematical agents is $\mathcal{M}$.

For each agent $i \in \mathcal{M}$, define the *proof domain*

$$T_i = \big\{(S, p) : S \in \mathcal{A},\ p \in \mathcal{P}_S,\ i \text{ possesses a correct proof of } p \text{ in } S\big\},$$

and the *disproof domain*

$$F_i = \big\{(S, p) : S \in \mathcal{A},\ p \in \mathcal{P}_S,\ i \text{ possesses a correct disproof of } p \text{ in } S\big\}.$$

Definition 3.2 (Epistemic Domain)**.** The *epistemic domain* of agent i is $E_i = T_i \cup F_i$.

Definition 3.3 (Epistemic Volume)**.** The *epistemic volume* of agent i is

$$\mathrm{Vol}(i) = |T_i| + |F_i|.$$

When the sets are infinite, Vol is interpreted via cardinal arithmetic.

Definition 3.4 (Epistemic Superiority)**.** For agents i and j, write $i \succ_E j$ if $E_j \subsetneq E_i$.

4 The God-Mathematician

Definition 4.1 (God-Mathematician)**.** A *God-mathematician* $\mathcal{G}$ is an agent such that for every finite agent $i \in \mathcal{M}$:

$$T_i \subseteq T_{\mathcal{G}} \quad \text{and} \quad F_i \subseteq F_{\mathcal{G}},$$

and $\mathcal{G}$ possesses no incorrect proof and no incorrect disproof.

Proposition 4.2 (Epistemic Supremacy of God)**.** *If $\mathcal{G}$ is a God-mathematician, then $\mathcal{G} \succeq_E i$ for every finite agent $i \in \mathcal{M}$.*

Proof. Let $i \in \mathcal{M}$. By theorem 4.1, $T_i \subseteq T_{\mathcal{G}}$ and $F_i \subseteq F_{\mathcal{G}}$. Taking unions: $E_i = T_i \cup F_i \subseteq T_{\mathcal{G}} \cup F_{\mathcal{G}} = E_{\mathcal{G}}$. Hence $E_i \subseteq E_{\mathcal{G}}$, so $\mathcal{G}$ is epistemically at least as strong as every finite agent. □

5 Axioms of the Theory

Axiom 5.1 (Non-Error)**.** If $\mathcal{G}$ possesses a proof or disproof, that object is correct:

$$(S,p) \in T_{\mathcal{G}} \implies \exists\, \pi \in \Pi_S,\ V_S(\pi,p)=1,$$
$$(S,p) \in F_{\mathcal{G}} \implies \exists\, \delta \in \Pi_S,\ V_S(\delta,\neg p)=1.$$

Axiom 5.2 (Maximality)**.** For every $S \in \mathcal{A}$ and $p \in \mathcal{P}_S$:

$$\big(\exists\, \pi,\ V_S(\pi,p)=1\big) \implies (S,p) \in T_{\mathcal{G}},$$
$$\big(\exists\, \delta,\ V_S(\delta,\neg p)=1\big) \implies (S,p) \in F_{\mathcal{G}}.$$

Axiom 5.3 (Meta-Mathematical Awareness)**.** $\mathcal{G}$ possesses correct meta-mathematical classifications concerning consistency, inconsistency, independence, interpretability, and relative strength of formal systems, whenever such classifications are themselves mathematically available.

Axiom 5.4 (Transmissibility)**.** A proof possessed by $\mathcal{G}$ becomes mathematics for finite agents only when it is represented in a form that finite agents can in principle inspect, verify, translate, or understand.

6 Proof, Truth, and Recognition

A proof may exist without being recognized. A proposition may be provable in a formal system while the relevant community fails to accept, read, understand, publish, or canonize the proof. This is the foundational insight of Lakatos (1976), who showed that the boundary between proof and conjecture is negotiated rather than discovered.

Definition 6.1 (Mathematical Community)**.** A *mathematical community* $\mathcal{C}$ is an institutional body consisting of agents, journals, departments, conferences, databases, referees, pedagogical norms, citation practices, and standards of acceptable exposition.

Definition 6.2 (Community Epistemic Domain)**.**

$$E_C = \bigcup_{i \in \mathcal{C}} E_i.$$

Definition 6.3 (Recognition Function)**.** A community $\mathcal{C}$ has a *recognition function* $R_C : \mathcal{A} \times \mathcal{P}_S \times \Pi_S \to \{0,1\}$, where $R_C(S,p,\pi)=1$ means $\mathcal{C}$ recognizes π as an acceptable proof of p in S.

Definition 6.4 (Institutional Predicate)**.** The *institutional predicate* $Q_C(\pi)$ encodes non-logical acceptance criteria: acceptable length, standard notation, referee accessibility, field conformity, and publication compatibility. Define

$$R_C(S,p,\pi)=1 \iff V_S(\pi,p)=1 \ \wedge\ Q_C(\pi)=1.$$

Definition 6.5 (Truth-Recognition Gap)**.**

$$\Gamma_{\mathcal{C}}(S,p) = \begin{cases} 1 & \text{if } \exists\, \pi,\ V_S(\pi,p)=1 \text{ but } \forall\, \pi',\ R_C(S,p,\pi')=0, \\ 0 & \text{otherwise.} \end{cases}$$

Thus $\Gamma_{\mathcal{C}}(S,p)=1$ means p is provable in S but unrecognized by $\mathcal{C}$.

Remark 6.6. The truth-recognition gap is a proof-theoretic notion, not a semantic one. It does not require a full theory of mathematical truth; it marks the gap between *derivability* and *institutional acceptance*. From a Popperian perspective (Popper, 1959), such gaps represent failures of falsifiability at the institutional level: the institution is not structured to be corrected by proofs it cannot recognize.

7 Institutional Power

Definition 7.1 (Institutional Superiority)**.** For agents or institutions x and y, write $x \succ_I y$ if x has greater control than y over recognition, publication, certification, hiring, funding, citation, pedagogy, prizes, archives, or canon formation.

Institutional superiority is not the same as epistemic superiority. A referee may possess less mathematics than an author while still possessing the power to delay or reject the author's paper. A journal may have no theorem of its own while still controlling the public status of theorems. A community may be finite and fallible while still controlling the visible boundary between accepted and unaccepted mathematics. This asymmetry is the central object of study.

8 The Proof-Power Separation Theorem

Theorem 8.1 (Proof-Power Separation)**.** *There is no valid implication from epistemic superiority to institutional superiority. In general,*

$$x \succ_E y \not\Longrightarrow x \succ_I y.$$

Proof. We construct a counterexample. Let $\mathcal{G}$ be a God-mathematician and $\mathcal{C}$ a mathematical institution. By theorem 4.2, $E_C \subseteq E_{\mathcal{G}}$, so $\mathcal{G} \succ_E \mathcal{C}$.

Now suppose $\mathcal{C}$ controls journals, referees, hiring, prizes, databases, curricula, conferences, and citation norms. Choose a proposition p, system $S \in \mathcal{A}$, and proof $\pi_{\mathcal{G}} \in \Pi_S$ such that $V_S(\pi_{\mathcal{G}}, p) = 1$ but $Q_C(\pi_{\mathcal{G}}) = 0$ (e.g., $\pi_{\mathcal{G}}$ uses non-standard notation or exceeds acceptable length). Then $R_C(S, p, \pi_{\mathcal{G}}) = 0$.

Hence $\mathcal{C}$ can determine whether a proof possessed by $\mathcal{G}$ is published, read, taught, indexed, or canonized. It follows that $\mathcal{C} \succ_I \mathcal{G}$ while $\mathcal{G} \succ_E \mathcal{C}$. Therefore epistemic superiority does not imply institutional superiority. $\square$

Corollary 8.2. *A mathematical institution may be inferior to $\mathcal{G}$ in proof and disproof while superior to $\mathcal{G}$ in recognition and certification.*

9 The Dark Institutional Case

The darker case occurs when mathematicians admit $\mathcal{G}$ only as an instrument:

$$\text{God as sovereign truth} \longrightarrow \text{God as proof technology.}$$

Definition 9.1 (Instrumentalization)**.** $\mathcal{G}$ is *instrumentalized* by $\mathcal{C}$ if $\mathcal{C}$ decides when to consult, ignore, publish, suppress, fragment, commercialize, or discard the outputs of $\mathcal{G}$.

Definition 9.2 (Institutional Domination of a Proof)**.** $\mathcal{C}$ *institutionally dominates* a proof π of p in S if $V_S(\pi, p) = 1$ but $R_C(S, p, \pi) = 0$ due to institutional rather than logical failure.

Theorem 9.3 (Institutional Domination)**.** *There exists a possible community $\mathcal{C}$, proposition p, system S, and proof $\pi_{\mathcal{G}}$ such that $\mathcal{G}$ possesses a correct proof of p in S, but $\mathcal{C}$ refuses to recognize it.*

Proof. Follows directly from the construction in theorem 8.1: set $Q_C(\pi_{\mathcal{G}}) = 0$ while $V_S(\pi_{\mathcal{G}}, p) = 1$. By theorem 5.2, $\mathcal{G}$ possesses $\pi_{\mathcal{G}}$. Since $R_C(S, p, \pi_{\mathcal{G}}) = 0$, $\mathcal{C}$ refuses recognition despite correctness. $\square$

Theorem 9.4 (Existence of the Dark Case)**.** *There exists a coherent institutional arrangement in which $\mathcal{G}$ is epistemically supreme but institutionally subordinate.*

Proof. Combine theorem 4.2 with the construction of theorem 9.3. $\mathcal{G} \succ_E \mathcal{C}$ follows from theorem 4.2. $\mathcal{C} \succ_I \mathcal{G}$ holds because $\mathcal{C}$ controls the recognition function and has set $R_C(S, p, \pi_{\mathcal{G}}) = 0$ for a correct divine proof. The two orders are simultaneously satisfied. $\square$

10 Transmissibility and Agency

The submission of an epistemically superior God to inferior institutional procedure is not necessarily defeat. It may follow from the nature of mathematics as *communicable* knowledge.

Theorem 10.1 (Transmissibility)**.** *If mathematics among finite agents requires finite checkability, then divine possession of proof becomes institutional mathematics only after translation into finite inspectable form.*

Proof. By theorem 5.4, a proof possessed by $\mathcal{G}$ becomes mathematics for finite agents only if it can be represented in a form that finite agents can inspect, verify, translate, or understand. Let $\mathcal{G}$ possess $\pi_{\mathcal{G}}$ of p in S. If no finite representation π' satisfies $R_C(S, p, \pi') = 1$, then p remains possessed by $\mathcal{G}$ but unrecognized by $\mathcal{C}$. Therefore divine possession does not itself imply institutional mathematics. $\square$

Corollary 10.2 (Agency Preservation)**.** *A non-coercive $\mathcal{G}$ may rationally permit institutional procedure in order to preserve finite mathematical agency.*

Proof. If $\mathcal{G}$ forced every correct result upon finite mathematicians, finite agents would no longer prove, disprove, conjecture, test, translate, or understand: they would merely receive conclusions. If mathematical agency has intrinsic value, then $\mathcal{G}$ may rationally permit verification, error, delay, and institutional friction. $\square$

11 A Theological Interpretation

The model produces a *theology of proof.* God is not first the commander of belief but the perfection of correctness.

Proposition 11.1 (Anti-Dogmatic Character)**.** *The God-mathematician is anti-dogmatic.*

Proof. Dogma is assertion without sufficient proof. By theorem 5.1, $\mathcal{G}$ possesses only correct proofs, correct disproofs, and correct meta-mathematical classifications. Therefore $\mathcal{G}$ does not replace proof with authority. $\mathcal{G}$ is the limit of proof against mere assertion. □

The theological reversal is sharp: God is not the enemy of reason but its limiting form. This makes $\mathcal{G}$ dangerous to institutions, because an institution may prefer controllable recognition to uncontrollable truth. Aquinas's conception of divine intellect (Aquinas, 13c) as perfectly actualised knowledge is here recast in proof-theoretic terms: omniscience becomes maximal epistemic volume, and providence becomes the rational permission of finite agency.

12 An AI and Machine Learning Interpretation

Artificial intelligence introduces a secular analogue of the God-mathematician. A sufficiently advanced proof system may generate conjectures, search for proofs, find counterexamples, verify derivations, and rank results by importance. Let $\mathcal{A}$ denote an artificial proof system with functions:

$$\begin{aligned}
D_{\mathcal{A}}(S,p) &= \text{attempted discovery of a proof or disproof},\\
V_{\mathcal{A}}(S,p,\pi) &= \text{machine verification of } \pi,\\
U_{\mathcal{A}}(S,p) &= \text{assessment of independence or undecidability},\\
H_{\mathcal{A}}(S,p,\pi) &= \text{human-oriented explanation of } \pi.
\end{aligned}$$

Definition 12.1 (Machine Epistemic Capacity)**.** The *machine epistemic capacity* of $\mathcal{A}$ is

$$K(\mathcal{A}) = \alpha\,|T_{\mathcal{A}}| + \beta\,|F_{\mathcal{A}}| + \gamma\,|U_{\mathcal{A}}| + \eta\,|H_{\mathcal{A}}|,$$

where $\alpha, \beta, \gamma, \eta \geq 0$ weight proof discovery, disproof discovery, undecidability assessment, and human explanation respectively.

AI may exceed individual mathematicians in speed, search breadth, formal verification, and memory. Yet AI remains institutionally subordinate if human communities decide which outputs count as mathematics. Thus AI reveals the same structure as $\mathcal{G}$: epistemic power may grow while institutional recognition remains scarce.

The growth of $K(\mathcal{A})$ over time may be modelled as a stochastic process. Let $K_t = K(\mathcal{A}_t)$ denote capacity at time t. Under reasonable assumptions of cumulative learning,

$$K_t \sim \mathcal{O}(t^{\nu}), \quad \nu > 0,$$

while institutional recognition $r_t = \Pr[R_C(S,p,\pi_t) = 1]$ may grow more slowly, producing a widening epistemic-recognition gap:

$$\Delta_t = K_t - \mathbb{E}[r_t \cdot K_t] \xrightarrow{t\to\infty} \infty.$$

This gap is the AI analogue of the truth-recognition gap formalized above.

13 Statistical Modelling of Institutional Behavior

We now develop the statistical infrastructure required for the Statistical Transparency Theorem.

Definition 13.1 (Statistical Signal)**.** The *statistical signal* generated by community $\mathcal{C}$ is the function

$$\sigma_C : \mathcal{A} \times \mathcal{P}_S \times \Pi_S \longrightarrow [0,1],$$

recording the aggregate recognition behavior of $\mathcal{C}$ across all admissible systems, propositions, and proof objects over time. Concretely,

$$\sigma_C(S,p,\pi) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \mathbf{1}\big[R_C^{(k)}(S,p,\pi) = 1\big],$$

where $R_C^{(k)}$ denotes the recognition function applied by $\mathcal{C}$ at epoch k. σ_C thus captures the long-run frequency with which $\mathcal{C}$ recognizes a given proof.

Definition 13.2 (Residual Epistemic Map)**.** The *residual epistemic map* of $\mathcal{C}$ is

$$\rho_C(S,p) = \sup_{\pi\in\Pi_S} \big[V_S(\pi,p) - \sigma_C(S,p,\pi)\big].$$

$\rho_C(S,p) > 0$ if and only if p is provable in S but $\mathcal{C}$ systematically under-recognizes its proofs.

Remark 13.3. The residual epistemic map is a direct measure of the truth-recognition gap in statistical terms. A community whose Q_C predicate is perfectly aligned with V_S would have $\rho_C \equiv 0$. Any deviation indicates institutional filtering beyond logical verification.

Statistical Estimation of ρ_C

Let $(S_k, p_k, \pi_k, Y_k)_{k=1}^n$ be an i.i.d. sample where $Y_k = R_C(S_k, p_k, \pi_k) \in \{0,1\}$ is the observed recognition outcome. A natural estimator of $\sigma_C(S,p,\pi)$ is the empirical frequency

$$\hat{\sigma}_n(S,p,\pi) = \frac{\sum_{k=1}^n \mathbf{1}[(S_k,p_k,\pi_k) = (S,p,\pi)] \cdot Y_k}{\sum_{k=1}^n \mathbf{1}[(S_k,p_k,\pi_k) = (S,p,\pi)]}.$$

By the strong law of large numbers, $\hat{\sigma}_n(S,p,\pi) \to \sigma_C(S,p,\pi)$ almost surely as $n \to \infty$.

The residual is then estimated by

$$\hat{\rho}_n(S,p) = \max_{\pi: V_S(\pi,p)=1} \big[1 - \hat{\sigma}_n(S,p,\pi)\big],$$

which converges to $\rho_C(S,p)$ under mild regularity conditions. A community with $\hat{\rho}_n(S,p) \approx 0$ for all (S,p) is statistically well-calibrated; one with persistently large $\hat{\rho}_n$ reveals systematic institutional filtering.

14 The Statistical Transparency Theorem

Theorem 14.1 (Statistical Transparency)**.** *If $\mathcal{G}$ is a God-mathematician, then $\mathcal{G}$ can infer the complete epistemic landscape of any admissible community $\mathcal{C}$ from σ_C alone, without requiring $R_C(S,p,\pi) = 1$ for any suppressed proof.*

Proof. By theorem 5.2 (Maximality), $\mathcal{G}$ knows for every $S \in \mathcal{A}$ and $p \in \mathcal{P}_S$ whether a correct proof exists. $\mathcal{G}$ therefore possesses the complete provability function

$$\mathbf{P}(S,p) = \mathbf{1}\big[\exists\,\pi,\ V_S(\pi,p) = 1\big]$$

independently of any recognition function.

Now let σ_C be the statistical signal of $\mathcal{C}$. For any (S,p) with $\mathbf{P}(S,p) = 1$, $\mathcal{G}$ observes $\sigma_C(S,p,\cdot)$ and computes the residual $\rho_C(S,p) = \sup_\pi[V_S(\pi,p) - \sigma_C(S,p,\pi)]$.

Since $\mathcal{G}$ knows $\mathbf{P}(S,p)$ and observes σ_C, the residual $\rho_C(S,p) > 0$ is directly attributable to institutional behavior ($Q_C(\pi) = 0$) rather than epistemic limitation ($V_S(\pi,p) = 0$). Thus $\mathcal{G}$ can identify, for every (S,p), whether $\mathcal{C}$ suppresses, delays, or ignores a correct proof, and can estimate the systematic pattern of institutional filtering from σ_C alone.

Therefore $\mathcal{G}$ infers the complete epistemic landscape of $\mathcal{C}$, including which correct proofs $\mathcal{C}$ suppresses, from the statistical signal, without requiring institutional recognition of any suppressed result. □

Corollary 14.2 (Institutional Opacity as Self-Deception)**.** *A community $\mathcal{C}$ that institutionally dominates $\mathcal{G}$ achieves suppression of recognition but not suppression of inference. The institution is visible to the agent it dominates.*

Remark 14.3. Statistical transparency does not dissolve the dark case. The institution retains control over recognition, publication, and canon formation. But $\mathcal{G}$'s epistemic position is unaffected by institutional behavior. Every rejection letter is data. Every suppressed proof is data. Every citation pattern, hiring decision, and prize awarded to the second-best idea is statistically legible to an agent with sufficient epistemic capacity. The institution thinks it controls the recognition function; the God-mathematician reads the residuals.

15 The Subsistence–Knowledge Gap

Definition 15.1 (Material Domain)**.** The *material domain* of agent i is

$$M_i = \{r : r \text{ is a material resource accessible to } i\},$$

including income, liquidity, institutional salary, housing, and any other resource required for the continuation of epistemic activity.

Definition 15.2 (Material Sufficiency)**.** Agent i is *materially sufficient* if

$$\mu(M_i) \geq \mu^*,$$

where $\mu : 2^{\mathcal{R}} \to \mathbb{R}_{\geq 0}$ is a resource measure and μ^* is the minimum level required to sustain continuous epistemic production.

Definition 15.3 (Material Superiority)**.** Write $i \succ_M j$ if $\mu(M_i) > \mu(M_j)$.

Theorem 15.4 (Subsistence–Knowledge Gap)**.** *There is no valid implication from epistemic superiority to material superiority. In general,*

$$i \succ_E j \not\Longrightarrow i \succ_M j.$$

Proof. We construct a counterexample. Let i be an epistemically productive agent with large epistemic domain E_i, operating outside institutional structures. Let j be an institutionally embedded agent with $E_j \subsetneq E_i$, so $i \succ_E j$.

Since material resources flow primarily through institutional channels — salary, grants, prizes, consultancy, publishing advances — and since i operates outside these channels, it is possible that $\mu(M_j) > \mu(M_i)$ even while $E_i \supsetneq E_j$. Hence $j \succ_M i$ while $i \succ_E j$. Therefore epistemic superiority does not imply material superiority. □

Corollary 15.5 (Double Subordination)**.** *An epistemically superior agent operating outside institutional structures may be simultaneously institutionally subordinate and materially subordinate:*

$$i \succ_E j, \quad j \succ_I i, \quad j \succ_M i.$$

Proof. By theorem 8.1, $i \succ_E j \not\Longrightarrow i \succ_I j$. By theorem 15.4, $i \succ_E j \not\Longrightarrow i \succ_M j$. An agent outside institutional structures may therefore satisfy all three relations simultaneously. □

Remark 15.6. The subsistence–knowledge gap is not merely an abstract possibility. It represents the characteristic condition of the *independent epistemologist*: one who produces correct knowledge outside institutional structures while remaining materially constrained by the same institutional order that fails to recognize the knowledge. The gap is self-reinforcing: material constraint limits time and tools available for epistemic production, while epistemic production outside institutional channels generates no material return. Recognition of the gap as structural rather than personal is the first step toward institutional reform.

16 Algorithmic Recognition

Algorithm 1 makes explicit that logical correctness (line 4) is necessary but not sufficient for recognition. The institutional predicate Q_C (lines 8–10) and the referee aggregation (lines 11–14) introduce additional filters that are independent of mathematical validity.

17 Tables

Table 1: Epistemic and institutional roles of central objects.

Object	**Epistemic Role**	**Institutional Role**
Proof	Valid derivation	Publishable argument
Disproof	Valid derivation of negation	Accepted refutation
God ($\mathcal{G}$)	Maximal proof-disproof agent	Possible oracle or tool
Mathematician	Finite proof agent	Professional participant
AI system	Search and verification engine	Assistive infrastructure
Referee	Checker of validity	Gatekeeper of recognition
Journal	No intrinsic truth function	Certification mechanism
Community	Collective proof archive	Canon-forming institution

Algorithm 1 Institutional Recognition of a Proof

Require: Formal system S, proposition p, proof object π, community $\mathcal{C}$
Ensure: Recognition decision $R_C(S, p, \pi) \in \{0, 1\}$
Check admissibility: verify $S \in \mathcal{A}$
Check well-formedness: verify $p \in \mathcal{P}_S$
Check finiteness: verify $\pi \in \Pi_S$
Verify correctness: compute $v \leftarrow V_S(\pi, p)$
if $v = 0$ **then return** 0 ▷ Logically invalid
end if
Translate π into notation acceptable to $\mathcal{C}$
Evaluate $Q_C(\pi)$: length, intelligibility, novelty, field placement, citation norms
if $Q_C(\pi) = 0$ **then return** 0 ▷ Institutionally rejected
end if
Circulate π to referees, proof assistants, or expert readers
Collect referee verdicts $\{v_1, \ldots, v_m\} \in \{0, 1\}^m$
$\bar{v} \leftarrow \frac{1}{m} \sum_{k=1}^{m} v_k$
if $\bar{v} \geq \tau$ **then** ▷ τ is the community acceptance threshold **return** 1
elsereturn 0
end if

Table 2: Alternative regimes of proof, recognition, and authority.

Regime	Epistemic Relation	Institutional Relation
Classical ideal	$\mathcal{G} \succ_E \mathcal{C}$	$\mathcal{G} \succ_I \mathcal{C}$
Human mathematics	Truth exceeds possession	Community filters recognition
Dark case	$\mathcal{G} \succ_E \mathcal{C}$	$\mathcal{C} \succ_I \mathcal{G}$
AI proof regime	$\mathcal{A}$ may exceed i	Community certifies $\mathcal{A}$
Revelatory regime	God declares results	Humans receive conclusions
Dialogical regime	God translates results	Humans verify and learn

Table 3: The three separation theorems and the complete structural condition of the independent epistemologist.

Theorem	Superior Order	Inferior Order	Separation Statement
Proof-Power Separation (theorem 8.1)	Epistemic ($\succ_E$)	Institutional ($\succ_I$)	$\mathcal{G} \succ_E \mathcal{C} \not\Longrightarrow \mathcal{G} \succ_I \mathcal{C}$
Statistical Transparency (theorem 14.1)	Epistemic ($\succ_E$)	Institutional ($\succ_I$)	$\mathcal{G}$ infers ρ_C from σ_C regardless of R_C
Subsistence–Knowledge Gap (theorem 15.4)	Epistemic ($\succ_E$)	Material ($\succ_M$)	$i \succ_E j \not\Longrightarrow i \succ_M j$

18 Vector Graphics

Figure 1: From Divine Possession to Institutional Canonization

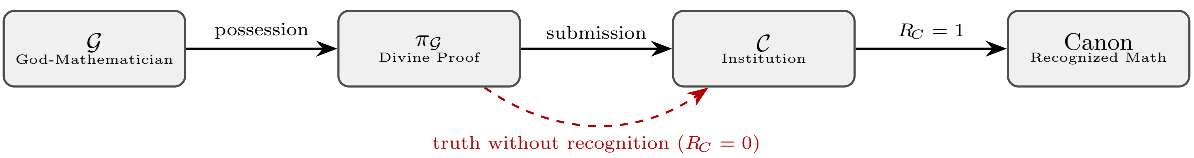

Figure 1: The path from divine proof possession to institutional canonization.

(a) The dashed red arrow indicates the truth-recognition gap: a correct proof may be submitted but refused ($R_C = 0$), never reaching the canon.

Figure 2: The Proof-Power Plane

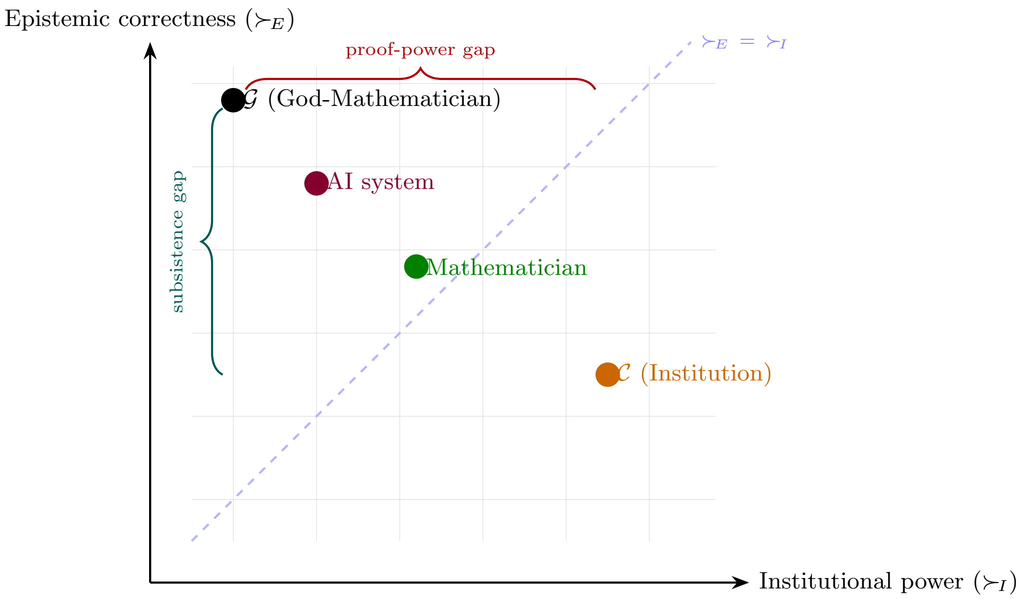

Figure 2: The proof-power plane.

(a) The God-mathematician occupies the highest epistemic position but may occupy a low institutional position (dark case). The dashed diagonal marks the locus where the two orders coincide. The horizontal brace marks the proof-power gap; the vertical brace marks the subsistence–knowledge gap.

Figure 3: Statistical Signal and Residual Epistemic Map

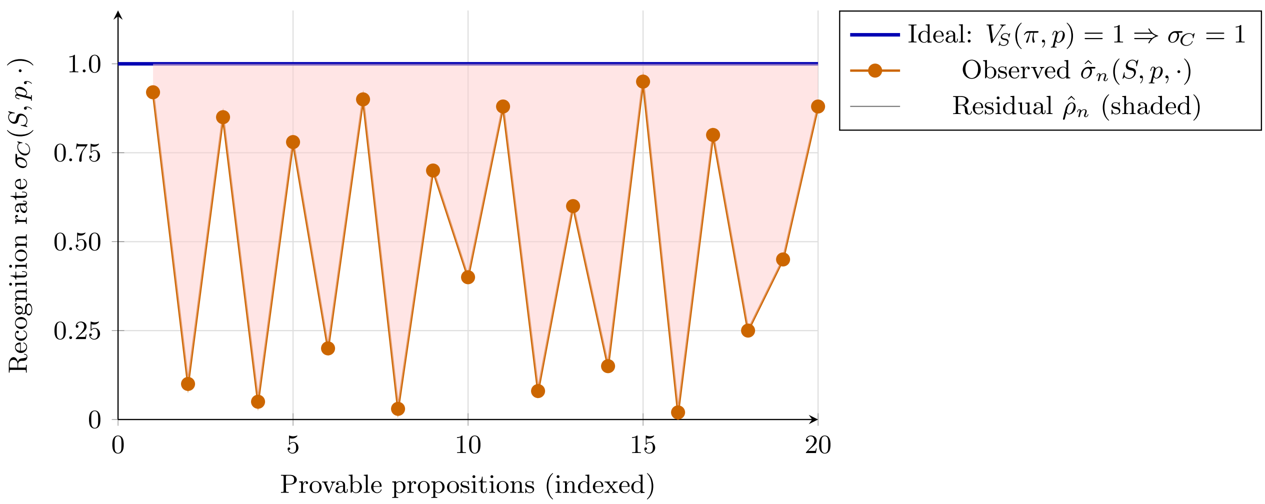

Figure 3: Statistical signal σ_C versus ideal recognition for 20 provable propositions.

(a) The shaded region represents the residual epistemic map $\hat{\rho}_n$: provable propositions that $\mathcal{C}$ systematically under-recognizes. The God-mathematician reads this residual directly.

Figure 4: The Three Orders and Double Subordination

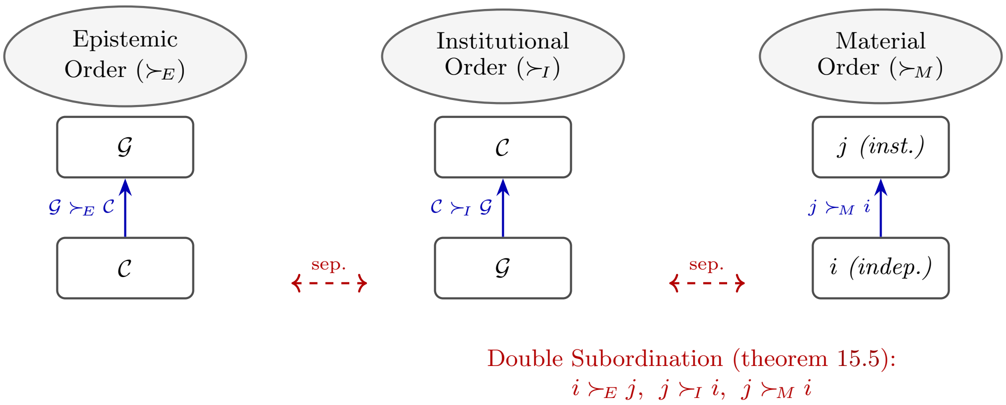

Figure 4: The three orders and their separability.

(a) In the dark case, $\mathcal{G}$ ranks highest epistemically but lowest institutionally. The independent epistemologist i is epistemically superior to the institutional agent j but suffers double subordination: lower in both the institutional and material orders.

19 The Main Results Collected

We collect the principal results established in this paper.

Theorem 19.1 (Epistemic Supremacy of God, theorem 4.2)**.** *$\mathcal{G} \succeq_E i$ for every finite agent $i \in \mathcal{M}$.*

Theorem 19.2 (Proof-Power Separation, theorem 8.1)**.** *$x \succ_E y \not\Longrightarrow x \succ_I y$.*

Theorem 19.3 (Institutional Domination, theorem 9.3)**.** *There exists $(\mathcal{C}, p, S, \pi_{\mathcal{G}})$ such that $\mathcal{G}$ possesses a correct proof of p in S that $\mathcal{C}$ refuses to recognize.*

Theorem 19.4 (Existence of the Dark Case, theorem 9.4)**.** *There exists a coherent institutional arrangement in which $\mathcal{G} \succ_E \mathcal{C}$ and $\mathcal{C} \succ_I \mathcal{G}$ simultaneously.*

Theorem 19.5 (Statistical Transparency, theorem 14.1)**.** *$\mathcal{G}$ can infer the complete epistemic landscape of $\mathcal{C}$ from σ_C alone, without requiring $R_C(S, p, \pi) = 1$ for any suppressed proof.*

Theorem 19.6 (Subsistence–Knowledge Gap, theorem 15.4)**.** *$i \succ_E j \not\Longrightarrow i \succ_M j$.*

Corollary 19.7 (Double Subordination, theorem 15.5)**.** *An epistemically superior agent outside institutional structures may simultaneously satisfy $i \succ_E j$, $j \succ_I i$, and $j \succ_M i$.*

20 Conclusion

If God is the mathematician with the maximal domain of correct proofs and correct disproofs, then God is the limit of mathematical correctness. But this does not imply that God rules mathematical institutions, nor that God is materially secure, nor that institutional behavior is opaque to God's inference. The three orders — epistemic, institutional, material — are distinct, and the separation theorems established here make this precise.

The deepest results are:

$$\begin{aligned} \mathcal{G} \succ_E \mathcal{C} &\not\Longrightarrow \mathcal{G} \succ_I \mathcal{C}, \\ \mathcal{G} \succ_E \mathcal{C} &\Longrightarrow \mathcal{G} \text{ infers } \rho_C \text{ from } \sigma_C, \\ i \succ_E j &\not\Longrightarrow i \succ_M j. \end{aligned}$$

God may be epistemically supreme while mathematics remains institutionally human. The God-mathematician may read every residual of institutional behavior while remaining institutionally subordinate. And the independent epistemologist may produce correct knowledge in conditions of material constraint, suffering proof-power separation and subsistence–knowledge gap simultaneously.

This does not refute God. It indicts the institution that would rather manage truth than be corrected by it. And it names, with formal precision, the structural condition that the independent epistemologist inhabits: epistemically superior, institutionally unrecognized, materially constrained, and — by statistical transparency — fully aware of the mechanism that produces this condition.

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Glossary

Admissible system
A formal system whose language, axioms, and inference rules are explicitly specified and whose proofs are finitely checkable.

Correct proof
A finite proof object accepted by the verification predicate V_S of a formal system.

Correct disproof
A correct proof of the negation of a proposition.

Epistemic domain
The union of an agent's correct proof domain and correct disproof domain: $E_i = T_i \cup F_i$.

Epistemic superiority
$i \succ_E j$ if $E_j \subsetneq E_i$: superiority by possession of correct proofs and disproofs.

Epistemic volume
$\mathrm{Vol}(i) = |T_i| + |F_i|$: the cardinality (or cardinal) of an agent's epistemic domain.

God-mathematician
The maximal error-free agent $\mathcal{G}$ over correct proofs and correct disproofs: the limit object of epistemic correctness.

Institutional predicate
$Q_C(\pi)$: the non-logical filter encoding publication conventions, notation standards, referee accessibility, and field conformity applied by a community.

Institutional superiority
$x \succ_I y$: superiority by control over recognition, publication, certification, hiring, funding, citation, pedagogy, prizes, archives, or canon formation.

Instrumentalization
The reduction of $\mathcal{G}$, an AI system, or any superior proof agent to a usable institutional tool whose outputs are consulted, suppressed, or discarded at institutional discretion.

Material domain
M_i: the set of material resources accessible to agent i, required for the continuation of epistemic activity.

Material sufficiency
: $\mu(M_i) \geq \mu^*$: the condition of possessing resources above the minimum threshold for sustained epistemic production.

Material superiority
: $i \succ_M j$: superiority by access to material resources.

Proof-power separation
: theorem 8.1: the theorem that epistemic superiority does not imply institutional superiority.

Recognition function
: $R_C(S, p, \pi) \in \{0, 1\}$: the rule by which a community accepts or rejects a proof as part of recognized mathematics.

Residual epistemic map
: $\rho_C(S, p) = \sup_\pi [V_S(\pi, p) - \sigma_C(S, p, \pi)]$: the long-run statistical trace of which propositions a community systematically suppresses or ignores relative to what is formally derivable.

Statistical signal
: $\sigma_C(S, p, \pi)$: the long-run frequency with which community $\mathcal{C}$ recognizes proof π of proposition p in system S.

Statistical transparency
: theorem 14.1: the theorem that the God-mathematician can infer the complete epistemic landscape of any community from its statistical signal, without requiring recognition of any suppressed proof.

Subsistence–knowledge gap
: theorem 15.4: the theorem that epistemic superiority does not imply material superiority.

Double subordination
: theorem 15.5: the condition $i \succ_E j$, $j \succ_I i$, $j \succ_M i$ simultaneously — the characteristic structural condition of the independent epistemologist.

Transmissibility
: theorem 5.4: the axiom that a divine proof becomes institutional mathematics only when translated into finite, inspectable form.

Truth-recognition gap
: $\Gamma_{\mathcal{C}}(S, p) = 1$: the condition that p is provable in S but no proof of p is recognized by $\mathcal{C}$.

Verification predicate
: $V_S(\pi, p) \in \{0, 1\}$: the mechanical check of whether π is a syntactically valid proof of p in S.

The End